

Project 1: Governance Artifacts & Audit Evidence

 **Portfolio Methodology Note:** This project profile represents a retrospective technical analysis of a real-world sovereign cloud deployment that I supported as an Advisor. While the physical implementation and code deployment were executed by the Core Systems Engineering division, this document demonstrates my technical aptitude to analyze, map, and architect the end-to-end security control logic of the environment through a professional Governance, Risk, and Compliance (GRC) lens.

Architectural Analysis Artifacts

The following artifacts are sanitized, illustrative functional-requirement models I authored to demonstrate how this sovereign cloud infrastructure's compliance objectives are systematically enforced. They represent the business logic I mapped, not production code.

Artifact A: Role-Based Access Control (RBAC) Matrix

Control Reference: NIST SP 800-53 Rev. 5 (AC-2: Account Management, AC-3: Access Enforcement)

Purpose: This configuration matrix maps system roles to explicit data access boundaries within the sovereign cloud enclave, preventing non-cleared commercial personnel from accessing federal system data.

System Role / Group	Data Path Access	SIEM Log Visibility	Infrastructure Configuration	Clearance Mandate Required
Federal Account Advisor	Read-Only (Account Tier)	Denied	Denied	US Citizen / Public Trust
Sovereign Support Engineer	Read/Write (Assigned Node)	Read-Only	Denied	US Citizen / T3 Secret Eligible
Partner Platform Lead	Full Enclave Access	Read/Write	Read-Only	US Citizen / T3 Secret Active
Core Systems Architect	Full Enclave Access	Full Access	Read/Write	US Citizen / T5 Top Secret

System Role / Group	Data Path Access	SIEM Log Visibility	Infrastructure Configuration	Clearance Mandate Required
Commercial Support (Default)	BLOCK_ALL	BLOCK_ALL	BLOCK_ALL	Non-Sovereign (No Access)

 Artifact B: Sovereign Boundary Access Protocol Blueprint

Control Reference: NIST SP 800-53 Rev. 5 (SC-7: Boundary Protection, SA-9: External System Services)

Purpose: Illustrative policy-logic outline describing the access-flagging perimeter logic when an external ticketing API interfaces with the sovereign database enclave. Presented as conceptual pseudocode, not a working system configuration.

CONCEPTUAL POLICY OUTLINE: IDENTITY INTAKE GATEWAY

FRAMEWORK VALIDATION TARGET: FEDRAMP MODERATE BASELINE

1. ROUTE INCOMING REQUEST

- If request originates from a public-sector fleet node, route to the isolated US sovereign cluster.
- Otherwise, route to the standard commercial path.

2. ASSESS IDENTITY

- Capture the user's authentication token.
- Cross-reference against the sovereign staff directory.
- If citizenship attribute is not US Citizen: deny access, log the event against NIST control AC-3, and terminate the session.
- Else if the user's role is not one of the approved sovereign roles: deny access and route a ticketing exception.

- Else: grant a role-scoped session over an encrypted tunnel.

Current Academic & Professional Progression

See the [top-level portfolio README](#) for certifications and credentials.